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Project 0, OpenMP Introduction

Thread Count	MegaMults/s	Speedup
1 Thread	2020.22	1
4 Threads	7092.64	3.510825554

The machine I ran this on was a desktop with an AMD 9700x using WSL. The graph above shows the performance between 1 thread and 4 threads. I ran the program around 10 times with 4 threads and half were around 7000, while the other half was around 5000. I assumed this has something to do with WSL, since my native Linux laptop gave me around 7000 as well.

The speedup between 1 thread and 4 threads is 3.51. We should see a speedup less than four because we are allocating 3 more threads that can process the array in parallel. It's not going to be exactly 4 because there is always some overhead and threads will most likely have to wait on each other for some processes.

The Parallel Fraction is 0.95.